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ABSTRACTS

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Anatomical and Histochemical Fingerprint of Gastrointestinal Tract in Bony Fishes from Different Trophic Levels

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Abstract

Anatomy, morphology and histology of gastrointestinal tract (GIT) show ontogeny and phylogeny variations that are an evolutionary adaption to feeding habit. The GIT includes the headgut (mouth and pharynx), foregut (esophagus, stomach and pyloric ceca), midgut (intestine), and hindgut (rectum) in fish. It has extensive endocrine functions, and plays important roles in food uptake and digestion, and nutrient uptake. In the present study, we aimed at anatomical and histochemical fingerprint of GIT in fish from different trophic levels. The GIT from omnivorous (tench, Tinca tinca and common bream, Abramis brama) and carnivorous (donkey croaker, Pennahia anea) fishes were fixed in 5% formalin. The GIT was sectioned and embedded in paraffin after dehydration. Serial sections of 7 µm thickness were stained with H&E or PAS. The histological slides were observed under a light microscopy, and images were captured for histochemical characterization and GIT morphometry. Unlike of T. tinca, stomach was present in A. brama and P. anea. Pyloric ceca were not observed in any species. For all species, GIT was composed of mucosa, submucosa, muscularis and serosa layers. Morphometry of mucosal folds and cellular component varied along GIT, and showed species variations. Gastric glands and goblet cells were common components of the stomach and postgastric region, respectively; higher numbers were observed at anterior stomach and posterior intestine. Histochemical traits were tightly aligned with feeding behavior. This study quantifies histological features of GIT, illustrates spatial distribution of cellular component along GIT, provides a framework for understanding trophic specialization, and expands insight into evolution of digestive system.

keywords: Gastric cells, Goblet cells, intestine, mucosal folds, stomach